Network IPS

Name: KhushiPatwa
Intern id:234

Meaning

A **Network Intrusion Prevention System (IPS)** is a security solution designed to monitor network traffic in real-time, identify potentially malicious activity or policy violations, and automatically take actions to block, prevent, or mitigate those threats. It acts as a proactive defense layer inline within the network path.

Objective

- **Detect** and **prevent** cyber threats and attacks such as malware, exploits, unauthorized access, and denial-of-service attacks.
- **Protect** network resources and critical assets from intrusion attempts.
- **Reduce** the risk and impact of attacks by stopping harmful traffic before it reaches vulnerable systems.
- **Provide visibility** and alerts for security teams to investigate suspicious activities.
- **Enforce network security policies** by controlling traffic flows.
-

Example

Suppose an attacker tries to exploit a known vulnerability in a web application by sending a malicious HTTP request with a suspicious URI like /admin.php?cmd=delete. The IPS:

- Inspects the incoming HTTP traffic.
- Matches the request pattern against known attack signatures.
- Drops the malicious packet.
- Logs an alert for the security team.

This prevents the attack from succeeding and protects the web server.

Types of Network IPS

Type	Description
Signature-Based IPS	Uses predefined signatures (patterns) of known attacks to detect threats. Effective against known threats but cannot detect new or unknown attacks.

Type	Description
Anomaly-Based IPS	Detects deviations from normal network behavior or traffic baselines. Can identify unknown threats but may have false positives.
Protocol-Based IPS	Monitors and enforces protocol compliance to detect protocol-specific attacks or misuse.
Behavior-Based IPS	Focuses on suspicious behavior, often using heuristics or machine learning to identify attacks.

Simple PoC Example Using Suricata

- Suricata running inline between external and internal networks.
- Rule configured to detect and drop HTTP GET requests containing /malicious.

Techniques Used in Network IPS

- **Signature Matching:** Comparing network traffic against a database of known attack patterns.
- **Protocol Analysis:** Verifying that traffic conforms to the expected rules of the protocol (e.g., HTTP, FTP).
- **Traffic Anomaly Detection:** Identifying unusual traffic patterns that deviate from baseline behavior.
- **Stateful Inspection:** Tracking the state of active connections and blocking packets that do not fit expected state transitions.
- **Rate Limiting:** Preventing Denial of Service (DoS) attacks by limiting traffic rate.
- **Deep Packet Inspection (DPI):** Examining packet payloads and headers beyond simple header inspection.

Summary

A Network IPS actively protects your network by inspecting traffic inline and blocking threats before they reach critical assets. It combines detection with prevention, making it essential for robust cybersecurity posture.